DEXTROSE 5%

Hypotonic once glucose used

Not for resuscitation

CHEMICAL NAME Glucose (D-glucose monohydrate) 50 g/L in water — 278 mmol/L glucose; ~200 kcal/L; isotonic in bag (~278 mOsm/L) but physiologically hypotonic ("free water")

CLASS Intravenous glucose solution (free water)

TRADE NAMES D5W · Glucose 5% infusion · Dextrose 5% in water

THERAPEUTIC USES / INDICATIONS

- Provide free water for hypernatraemia and water deficits.
- Maintenance fluid component (with saline and K^+) — avoid as sole maintenance in children.
- Diluent for drugs requiring glucose (amiodarone, amphotericin, oxaliplatin, noradrenaline in some protocols).
- Prevent hypoglycaemia during fasting (with insulin infusions; VRIII).

ADMINISTRATION

- **IV** peripheral line; rate by prescription.
- Never for fluid resuscitation (distributes into all body water).
- Monitor glucose in diabetics.

DOSAGES

- **Hypernatraemia:** calculated free-water deficit; lower Na^+ slowly in chronic cases, with frequent Na^+ checks.
- **With VRIII:** as substrate fluid per local protocol (e.g. 0.45% saline + 5% glucose + KCl).
- Rate per prescription; adult maintenance water 25–30 mL/kg/day in total.

PHARMACOKINETICS

- Glucose metabolised → leaves free water that distributes through total body water (only ~8% stays intravascular).

PHARMACODYNAMICS

- Provides **water and calories** (not enough for nutrition); dilutes serum sodium.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Phlebitis.
- Hyperglycaemia (diabetics, stress).

ADVERSE EFFECTS (SERIOUS)

- Hyponatraemia (especially post-operative patients and children with ADH release) → cerebral oedema, seizures, death.
- Fluid overload; hypokalaemia (insulin release).
- Wernicke's encephalopathy if thiamine-deficient.

CONTRAINDICATIONS & CAUTIONS

- Hyponatraemia; raised ICP/head injury, stroke (hypotonic fluids worsen cerebral oedema).
- Hyperglycaemia; resuscitation of hypovolaemic shock.

KEY INTERACTIONS

- Incompatible with some drugs (e.g. phenytoin, ceftriaxone in some combinations) — check before mixing.

NURSING MANAGEMENT

- Monitor serum Na^+, glucose, fluid balance.
- Avoid as sole maintenance in children — isotonic fluids preferred.
- Thiamine before glucose in at-risk patients.
- Watch for headache, nausea, confusion (hyponatraemia).

PATIENT EDUCATION

- Report headache, confusion, nausea or swelling.

KEY POINTS

- D5W; essentially free water.
- Risk of hyponatraemia (children, post-op).
- Treats hypernatraemia; drug diluent.
- Never for resuscitation or head injury.

SOURCES CHECKED **[958]** Glucose 5 % w/v Intravenous Infusion BP (UK SmPC) **[954]** NICE CG174

Full references p. 432–457